Self-Certification Program

Guidelines for Kubernetes Distributions

This document outlines the process for Independent Software Vendors (ISVs) to certify their Kubernetes distribution for interoperability with Run:ai.

Program Objective

The **NVIDIA Run:ai Self-Certification Program** enables ISV partners to independently validate their Kubernetes distribution's compatibility with the Run:ai platform. Certification covers core Kubernetes API compatibility, GPU scheduling, workload execution, and custom resources.

Pre-Certification Installation Requirements

Before starting the certification process, install the required NVIDIA Run:ai components according to the official documentation:

- **NVIDIA Run:ai Control Plane:**
  Deploy the control plane to manage your cluster(s). The control plane and clusters require Kubernetes. Typically, the control plane and the first managed cluster are deployed on the same Kubernetes cluster.

- **One or More Clusters:**
  Set up one or more clusters as described in the NVIDIA Run:ai documentation
  For installation details and system/network requirements, refer to:
  NVIDIA Run:ai Installation Documentation
    - Follow the steps for your environment (Self-hosted, connected)
    - Use the provided Helm charts and follow permissions and configuration guidance.

Certification Process Overview

- **Access Toolkit & License:** Upon approval, receive program documentation, the automated certification toolkit (Docker image), and a license for NVIDIA Run:ai.
- **Prepare Environment:** Set up a test Kubernetes cluster with the NVIDIA Run:ai control plane and cluster(s) installed.
- **Run Validation:** Deploy the toolkit in your cluster environment; only kubeconfig access and minimal privileges required.
- **Submit Report:** Upon completion, submit the structured test results.
  The results archive is located in the `results` folder with the filename:

  `runai-certification-results-<timestamp>.tar.gz`

  Example:
  `results/runai-certification-results-20250918143000.tar.gz`
- **Review & Publish:** NVIDIA Run:ai will review submissions, approve certifications, and publish certification levels in official documentation and on the partner portal.

Technical Instructions

The certification toolkit is provided as a **Docker image**.
Registry details will be supplied upon entry into the program

Load the Docker Image

```
None

# Load the AMD64 certification kit image

docker load -i certification-kit-amd64.tar.gz
```

```
# Load the ARM64 certification kit image

docker load -i certification-kit-arm64.tar.gz
```

Run the Kit and Generate Results

```
None

mkdir -p results && \

docker run --rm \

-e CONTROL_PLANE_URL=<CONTROL_PLANE_URL> \

-e CLUSTER_NAME_PREFIX=<CLUSTER_NAME_PREFIX> \

-e CLUSTER_URL=<CLUSTER_URL> \

-e CONTROL_PLANE_ADMIN_USERNAME=<ADMIN_USERNAME> \

-e CONTROL_PLANE_ADMIN_PASSWORD='<ADMIN_PASSWORD>' \

-v $(pwd)/kubeconfig:/kubeconfig/config:ro \

-v $(pwd)/results:/app/e2e/results \

runai/certification-kit-amd64:latest && \

cd results/allure-report && \

npx http-server -p 8080 & \

sleep 3 && \

open http://localhost:8080
```

Execution Example

```
None

mkdir -p results && \

docker run --rm \

-e CONTROL_PLANE_URL=https://cert-kit-demo.runailabs.com \

-e CLUSTER_NAME_PREFIX=cert-kit-demo \

-e CLUSTER_URL=https://cert-kit-demo.runailabs.com \

-e CONTROL_PLANE_ADMIN_USERNAME=test@run.ai \

-e CONTROL_PLANE_ADMIN_PASSWORD='Abcd!234' \

-v $(pwd)/kubeconfig:/kubeconfig/config:ro \

-v $(pwd)/results:/app/e2e/results \

runai/certification-kit-arm64:latest && \

cd results/allure-report && \

npx http-server -p 8080 & \

sleep 3 && \

open http://localhost:8080
```

Note: Make sure to place your kubeconfig file (containing the cluster kubeconfig details) in the same directory from which you run this command.

This command runs the toolkit interactively, saving all test outputs in the local results folder.

The process will prompt for the Run:ai control-plane URL, cluster name, cluster URL, and kubeconfig, all of which are required to execute the certification and generate results for submission.

This launches a lightweight local web server and opens the generated report in the browser for review.

The Certification Kit automatically creates a package at the end of each test run, containing all necessary files for Run:ai to review results.
This package is immediately available for sharing with the Run:ai team as part of the certification submission process.

Certified Versions

Certification is performed against the latest Run:ai General Availability (GA) release, which is issued every three months.
Each certification must be completed for every unique combination of:

- Run:ai Version
- Kubernetes Version
- Your Distribution Version

Partners are required to maintain their certification status by validating each new combination promptly after a new GA release.

When a new Run:ai GA, Kubernetes version, or distribution version is released, certification must be repeated to ensure continued compatibility.

Support Process

All self-certified distributions are supported directly by the **partner organization**.
Partners are responsible for:

  - Investigating and resolving issues specific to their Kubernetes distribution.
  - Ensuring ongoing compatibility with new Run:AI releases.
  - Maintaining up-to-date documentation for their users.

2. **Customer Support Flow**
   If a customer encounters an issue on a self-certified distribution:
   - The customer should **contact the partner's support team first**.
     If the issue is determined to be related to the Run.ai platform (not the distribution or Kubernetes), the cutsomer may escalate it to NVIDIA via standard partner support channels.
   - NVIDIA support will collaborate with the customer to resolve Run:ai-specific issues but will redirect Kubernetes-related issues back to

the partner.

3. **Ongoing Validation**
    - Partners are expected to revalidate their distribution with each major Run:ai release to maintain their self-certified status.

Please find the docker image in this link.
In order to get access to the docker image, send the results or for any questions, please contact:
isv-certification-run.ai@nvidia.com